EXPERIMENT 9 – STUDENT OBSERVATION

1. Important features of Wireless Router and DHCP Server configuration

Wireless Router: A wireless router enables devices to connect to a wireless network. It can be configured with a network name (SSID), for example, HomeNetwork. It may also provide DHCP services and use a specified DNS server address.

DHCP Server: A DHCP server automatically provides network devices with IP addresses, subnet masks, default gateways, and DNS details. This avoids configuring each device manually.

2. Importance of a DHCP Server in internetworking

DHCP automatically supplies the network settings required by connected devices. It minimizes manual setup, helps reduce IP address conflicts, and simplifies network management. When a device joins the network, it can receive its configuration without the user entering each value.

3. Design and configure an internetwork using switches, a router, and Ethernet cables

Connect the devices in the following order: PC0 → Switch → Router → Switch → PC1. Use copper straight-through Ethernet cables for the connections.

Device / Interface	IP Address	Subnet Mask	Default Gateway
Router Fa0/0	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
Router Fa0/1	192.168.20.1	255.255.255.0	—
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Check the connection by using the **ping** command or by sending a Simple PDU from PC0 to PC1.

Note: The uploaded lab-manual text places these questions under Practical 10. This document keeps the requested title, Experiment 9.